

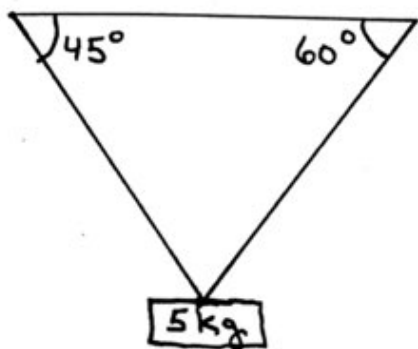
Force as a Vector

Example 1

A force of 20N acts west and a force of 10N acts north on a particle. Find the direction and magnitude of the resultant and the equilibrant of these 2 forces.

Example 2

A traffic sign with a mass of 5kg is suspended above the street as shown in the diagram below. Find the tension (force) in each of these wires.



Forces

- 1) Find the resultant and equilibrant of forces of 33N and 47N acting on a point at an angle of 40° to each other.

 - 2) A 3kg metal bar is suspended from the middle of a 2m chain whose ends are attached to a support beam 1.5 m apart. Find the tensions in each part of the chain.

 - 3) A box with mass of 12kg rests on a frictionless ramp inclined at an angle of 25° to the level ground. Find the magnitude of the force parallel to the ramp and of the force perpendicular to the ramp.

 - 4) Use vectors to show that the points A(1, 11), B(-2, -4) and C(5, 31) are collinear points.
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